

CookingSteps – Learn International Dishes Step by Step

1

Team: SALAMA, ZHAMANKHAN and ZHETESSOV
M2 Computer Science – University Paris-Saclay

Overview

- Situation & Problem
- Solution & Value Proposition
- Functionalities & Screens
- Acquisition & Retention
- Market Study
- Detailed Analysis of Users' Behavior
- Users Problem and App

Situation and problem

► Situation & User:

Nur, 23 years old, novice home cook, wants to prepare international dishes such as Indian butter chicken or Palestinian *maqlouba*.

► Problem:

Online recipes contradict each other and often skip essential technique steps. Beginners cannot know which version reflects real home cooking in the dish's culture. This leads to failed attempts, wasted ingredients, and loss of confidence.

Solution and value proposition

➤ **Solution:**

A mobile app providing step-by-step cooking guidance for international dishes, created directly by people who grew up cooking them. Each recipe includes clear photos/videos, precise timing, real texture/heat instructions, and small cultural notes.

➤ **Value Proposition:**

Reliable, culturally faithful instructions that help beginners cook dishes successfully on the first try.

Validated by users:

- 100% found navigation easy
- 100% understood steps clearly
- 100% used images to follow technique
- 90% successfully cooked the dish
- 90% satisfied (avg rating 4.7/5)
- 90% would recommend the app
- 83–90% would cook again with it

Functionalities and screens

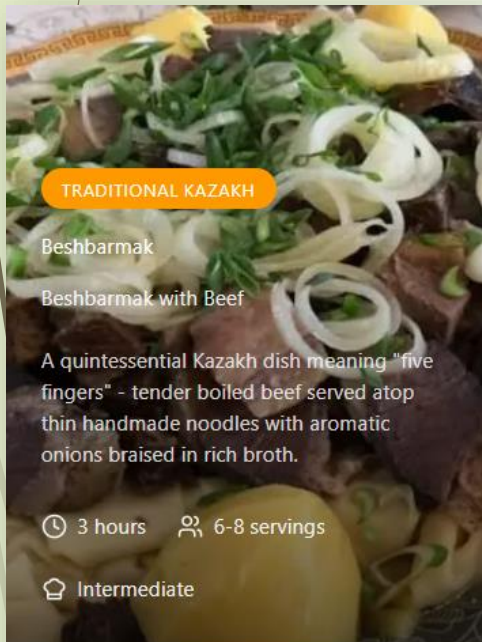
Features

- No registration/login
- When user opens app:
 - User selects a dish to cook
 - Ingredient list
- Step-by-Step Guidance
 - First step in the cooking process is displayed clearly
 - Next step shown by swiping screen down
 - Cultural tips at the end
- Timer per step (CountDownTimer)
- Push notifications suggesting new dishes

Future features

- Unlock achievement badges for milestones
- Rating and Comments allow to get user feedback

CookingSteps traditions kept!



Ingredients

For the Broth

- 1.5 kg — beef (bone-in for best flavor)
- 3 pcs — onions
- 1 pc — carrot
- 3-4 L — water
- to taste — salt

For the Dough

- 500 g — flour
- 1 pc — egg
- 1 tsp — salt
- 200 ml — water

For Serving

- 1 bunch — green onion (chopped)



Step-by-Step Instructions



Prepare the Broth

Place the beef, one onion (whole), and carrot in a large pot. Fill with cold water (3-4 liters) to cover the meat completely.



Skim the Foam

Bring to a boil over medium-high heat. As the broth starts to boil, skim off the foam that rises to the surface with a spoon.

CR: Acquisition and Retention – Strategy, Accounting and Analysis

➤ **Acquisition – QR codes:**

- Flyers with unique QR codes

Accounting: scan count + timestamp

Analysis: location performance + best time slots

➤ **Acquisition – Word of mouth:**

- Personal sharing links

Accounting: unique QR code per user

Analysis: compare family/friends/student clubs

➤ **Retention – Push notifications:**

Accounting: delivery + open + interaction rates

Analysis: opened → higher reuse intention (83–90%)

➤ **Retention – Achievement system:**

Accounting: badge unlock logs

Analysis: badge users show stronger completion (90%)

Market Study: Users/Customers – Qualitative & Quantitative

Qualitative

- 100% found navigation to be easy
- 100% understood steps clarity, images, and completed at least one recipe
- 83% will use app again

Quantitative

- 68 736 000 France population
- 33 652 015 From 20 to 59 years
- 94% Smartphone ownership
- 20 minutes daily time in apps
- 68% Loves cooking
- 40% Use recipe website apps
- "recipe" 75 searches/daily in France

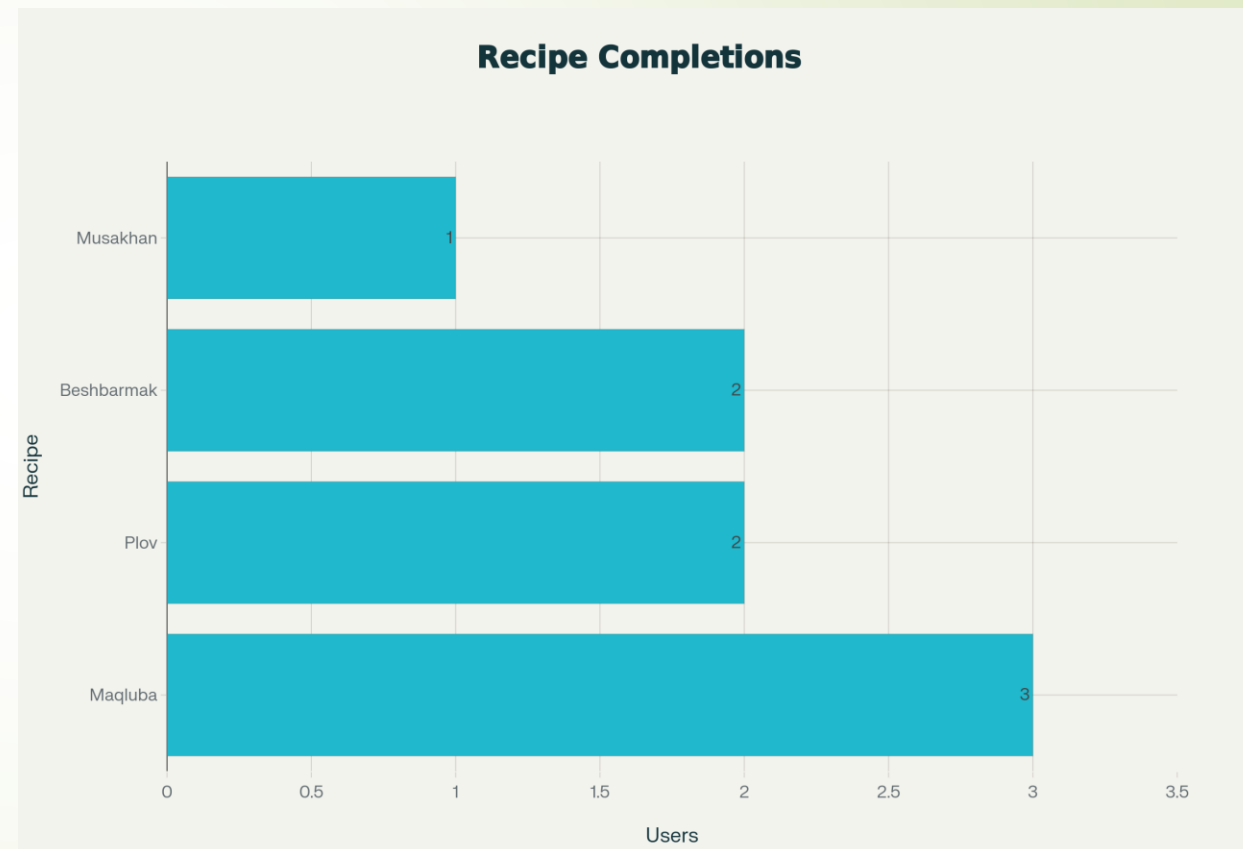
Market Study: Competitors – Qualitative & Quantitative

Feature	CookingSteps	YouTube	AllRecipes	Yummly	MasterClass	Tasty
Solves recipe clarity	✓ Yes	Partial	Partial	Partial	✓ Yes	✓ Yes
Step-by-step guidance	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes
Cultural authenticity focus	✓ USP	X No	X No	X No	X No	X No
Beginner-friendly instructions	✓ Yes	Partial	✓ Yes	✓ Yes	✓ Yes	✓ Yes
Free tier available	✓ Yes	✓ Yes	✓ Yes	Partial	X No	X No
Premium subscription	Planned	X No	Partial	✓ Yes	✓ Yes	X No
Native chef/cultural experts	✓ Core	X Mixed quality	X User-generated	X No	✓ Yes	X No
Targeted at intl cuisines	✓ Focus	X Generic	X Generic	X Generic	X No	✓ Viral trends
Mobile app available	✓ Yes	✓ Yes	✓ Yes	✓ Yes	Partial	✓ Yes
Community/user reviews	Planned	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes

Detailed Analysis of Users' Behavior w.r.t. the Problem

9.1. one week after the deployment of the app

- 3 peak daily active users, 8 total users
- 100% recipe completion rate (8/8 users rated their dishes)
- 4.8/5 average satisfaction rating
- 100% feature adoption (step-by-step guidance), 38% returning users
- 90% cooking success, 83-90% reuse intention



Users Problem and App: Conclusion and Decision Regarding the Project

GO DECISION - CONTINUE PROJECT

Problem Solved

Recipes now clear
Native authenticity
90% cooking success
User confidence ↑ up

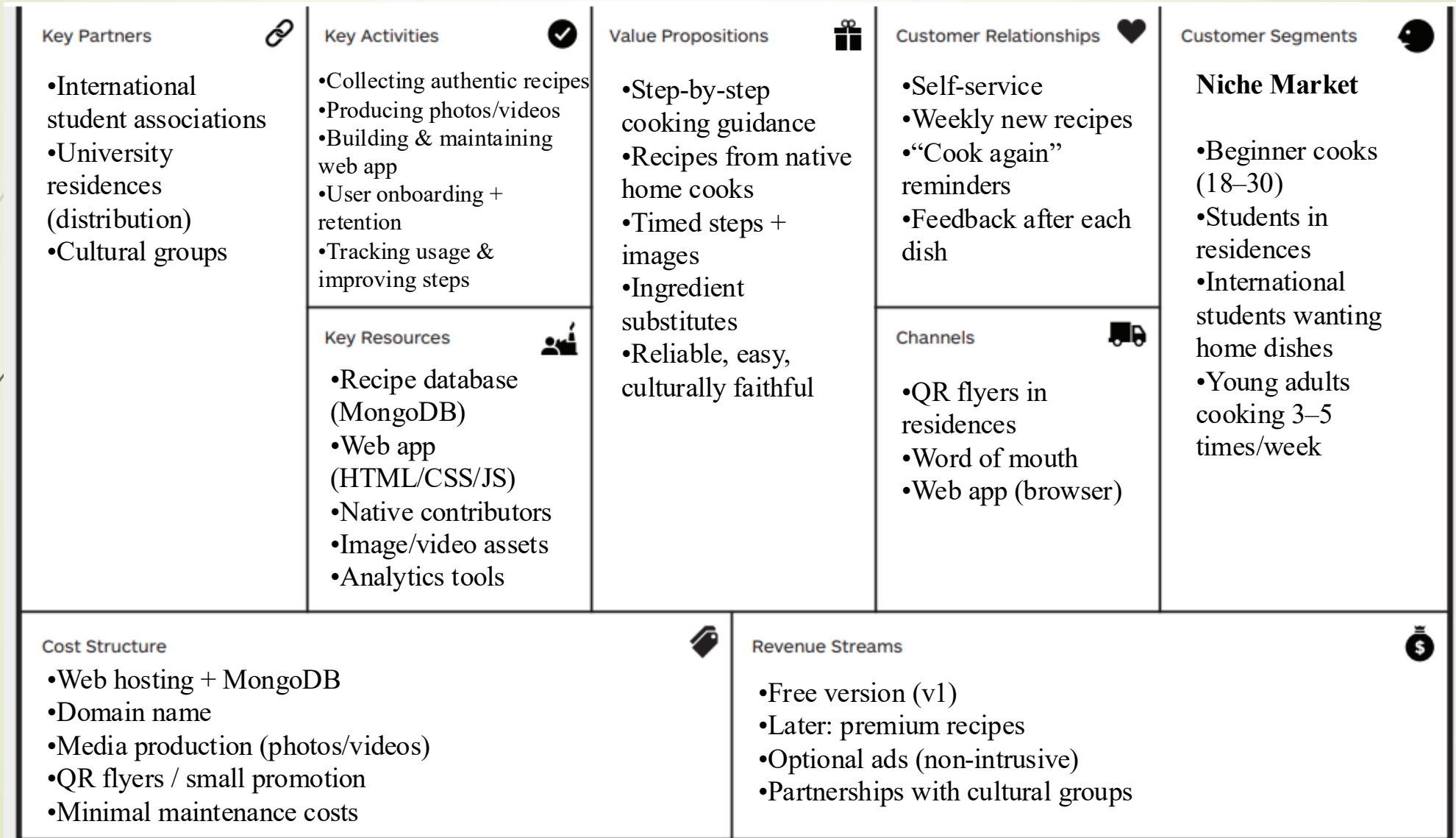
Retention & NPS
38% returning users
4.8/5 rating
90% recommend
83-90% reuse intent



2b. Appendix Overview

- BMC
- Java code
- Manual prototype
- Application tests
- Users list (acquisition/retention)
- Interviews & questionnaires
- Market studies
- Feedback analysis
- Financial plan

3b. BMC, including revenue streams, price



5b. Java code + Demonstration

```
package com.example.beshbarmak;

import android.os.Bundle;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```


5c. Manual Prototype: Detailed Design

- **Goal:** simulate app behavior without coding
Tools: printed recipe steps, notebook, phone timer, SMS
- **Data:** ingredients list, steps, durations, cultural notes stored in notebook (app database simulation)
- **Process:**
 - User chooses dish from printed menu
 - Follows printed step-by-step sheet
 - Team member triggers timers manually
 - SMS sent when each step's timer ends
 - Photos/videos replaced by small printed images
- **Purpose:** validate clarity, timing, user understanding before coding

5d Application (Website/Mobile App): Technology Chosen, Team Tests and Team/External Interface Tests

➤ 5d.1 – Technology Chosen (Website)

- **Platform:** Responsive web application
- **Language:** HTML / CSS / JavaScript
- **Data storage:** MongoDB (recipes, steps, contributors)
- **Media:** Local images / short clips
- **Timers:** JavaScript countdown
- **Notifications:** Browser notifications API

Why: scalable, fast to build, JSON-friendly, accessible on all devices

➤ 5d.2 — Team Tests (Internal Tests)

- Load recipes from MongoDB
- Display recipe list (HTML table / cards)
- Render step-by-step page correctly
- Display images/clips reliably
- Navigation between screens
- Check for broken steps or missing fields
- Internal usability test (clarity, readability)

➤ 5d.3 — External Interface Tests (With Users)

- 10 students test recipe flow
- Tasks: choose dish → follow steps → complete recipe
- Observe confusion points (timing, unclear step, missing photo)
- Test on multiple devices (phone/laptop/tablet)
- External feedback on clarity + cultural notes

6b. CR: Acquisition & Rétention – Users List

- Arsalan
- Menghor
- Lina
- Saja
- Niko
- Tyler
- Victor
- Farah
- Jad
- Lilian

7b. Interviews: Questions & Results

- Users interviewed: 10 students, beginner cooks, international + local
- **Questions:**
 - Interface clarity?
 - Navigation smooth?
 - Steps clear?
 - Images helpful?
 - Success cooking?
 - Reuse intention?
 - Improvements?
- **Results:**
 - Interface: 9/10 “easy” (avg 4.7/5)
 - Navigation: 9 “yes”, 1 “mostly”
 - Steps clear: 9/10 (avg 4.6/5)
 - Images helpful: 9/10
 - Success: 9 “yes”, 1 “almost”
 - Reuse: 9 “yes”, 1 “maybe”
 - Requests: more recipes, better photos

7c. Questionnaire: Questions & Results

- **Respondents:** 10 students (5 beginners, 5 intermediate)
- **Key results:**
 - Usage:** 5 once, 5 used 2–3×
 - Interface clarity:** 90%
 - Navigation:** 90% smooth
 - Steps clarity:** 90%
 - Images helpful:** 90%
 - Cooking success:** 90%
 - Reuse intention:** 90%
 - Recommend:** 90%
 - Overall rating:** 4.7/5
 - Main motivations:** more recipes (50%), better photos (30%), saving history (10%)

7d. Market Study (Users/Customers) – Sources & Additional Elements

- Statista.com
 - Meaning of cooking for people in France in 2024
 - Do French people like cooking?
 - Daily time spent on mobile apps in France in October 2022, by age and gender
 - Smartphone penetration in France in 2023, by age group
 - Share of internet users who get meal inspiration from various online sources worldwide as of November 2021, by generation
- Google Trends
 - "recipe" web search in France
- Interview data
- Questionnaire

8b. Market Study (Competitors) – Sources & Additional Elements



Competitor	Price	France Users	Rating	Key Metric
YouTube	Free (ads)	45M+	N/A	2.1B searches/yr for recipes
AllRecipes	Free+€2-3/mo	500K-1M	4.2/5	65,000 recipes
Yummly	Free+€4.99/mo	300K-500K	4.3/5	15M recipes (AI-sorted)
MasterClass	€100-200/yr	50K-100K	4.6/5	Expert-led courses
Tasty	Free (ads)	150K ↓	3.5/5	Declining 3-min videos
CookingSteps	Free+€1-2/mo	Target 18-30	In dev	Authentic step-by-step

9b. User Behavior & Feedback – Manual Prototype (Team & External)

- **Setup:** printed recipe steps + notebook data + manual timers (SMS/phone)
Users: 4 testers (students, beginner cooks)

- **Observed behavior:**
 - Users read steps carefully
 - Confusion on heat level (medium/high?)
 - Users wanted “Next step” confirmation
 - Ingredient substitutes requested frequently
 - Cultural notes appreciated (2/4 asked for more)

- **Feedback summary:**
 - Step structure OK, but visuals needed
 - Strong interest in culturally faithful instructions

9c. User Behavior & Feedback – Application (Team & External)

➤ **Observed usage:**

- Most users followed steps sequentially
- Timers used consistently
- Images viewed often for key steps
- Cultural notes read by motivated users
- No navigation blockers

➤ **Feedback:**

- Very high clarity (avg 4.6–4.7/5)
- 90% completed dish successfully
- Strong retention (90% reuse intention)
- Strong recommendation rate (90%)
- Requests: more recipes, optional history

10b. Financial Plan (Limited): Expenses, Break-even Arguments

Metric	Year 1	Year 2	Year 3
Paying Users (end)	60	240	360
ARPU (€/month)	€ 2.50	€ 2.80	€ 3.00
Annual Revenue	€ 900	€ 5,040	€ 10,800
Monthly OpEx	€ 150	€ 300	€ 500
Annual Profit/Loss	-€900	€ 1,440	€ 4,800
Gross Margin %	-100%	29%	44%
Status	Investment	Breakeven	Profitable

Bootstrap Model: No salaries initially, minimal infrastructure

Revenue: Free version + €2.5–3/mo premium + optional ads. Growth: Organic user acquisition, word-of-mouth (zero CAC), student partnerships.
Unit economics: CAC €100 → LTV €300–600 (3–6x ratio). Reinvestment: Profits from Year 2 fund product expansion & team